A 2-D Unstructured Finite-Volume Compressible Navier–Stokes Solver with MPI Partitioning and Implicit Integration

cfd2d implementation

October 24, 2026

Abstract

This report documents an original C++17/MPI finite-volume solver for the two-dimensional compressible Navier–Stokes equations on the supplied unstructured CGNS meshes. The solver uses mixed triangle/quadrangle cells, cell-centred gradients, limited piecewise-linear reconstruction, Roe fluxes with an entropy fix, gradient-based Newtonian viscous fluxes, and a block LU-SGS implicit pseudo-time solver. METIS partitions the cell graph; each rank stores only owned cells, halo cells, and local faces. A first-order implicit startup is followed by the limited second-order production stage. Results, convergence evidence, forces, field visualizations, the Re=200 vortex-street analysis, and limitations are reported for all required cases.

Contents

1 Governing equations **1**

2 Mesh, geometry, and boundary conditions **2**

3 Finite-volume discretization **2**

4 Implicit integration and MPI implementation **2**

5 Production run summary **3**

6 Rank-count consistency **3**

7 cylinder_m010_laminar_re20 **4**

8 cylinder_m010_laminar_re200 **4**

9 naca0012_m015_inviscid **6**

10 naca0012_m015_laminar_re5000 **6**

11 naca0012_m080_inviscid **7**

12 naca0012_m080_laminar_re5000 **7**

13 naca0012_m200_inviscid **7**

14 naca0012_m200_laminar_re5000 **9**

15 Validation and limitations **9**

16 Reproducibility **11**

17 Extensibility and conclusion **11**

1 Governing equations

The conservative state is

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T. \tag{1}$$

The compressible Navier–Stokes equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{2}$$

with

$$\mathbf{F}^i = [\rho u,\ \rho u^2 + p,\ \rho uv,\ u(\rho E + p)]^T, \tag{3}$$

$$\mathbf{G}^i = [\rho v,\ \rho uv,\ \rho v^2 + p,\ v(\rho E + p)]^T. \tag{4}$$

For a calorically perfect gas,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \tag{5}$$

with $\gamma = 1.4$, $R = 1$, and $Pr = 0.72$. The viscous stresses are

$$\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \tag{6}$$

and the heat flux is $\mathbf{q} = -k\nabla T$ with $k = \mu c_p/Pr$. The constant laminar viscosity is $\mu = \rho_\infty U_\infty L_{\text{ref}}/Re$. All quantities are nondimensional; the freestream density and velocity magnitude are unity, and the pressure follows from the Mach number.

2 Mesh, geometry, and boundary conditions

The CGNS reader handles mixed unstructured zones and abutting one-to-one interfaces. Interface vertices are merged by identity, duplicate faces are removed, and boundary sections are mapped to the case-file families (`bc-2`, `bc-4`, `WALL`, `FAR`). Cell volumes and centroids are computed by exact decomposition of triangles and quadrilaterals. Face normals are oriented outward from the left cell. The NACA mesh has 20,816 cells and 36,498 unique faces; the two-zone cylinder mesh has 10,185 cells, 20,180 faces and 9,995 merged nodes.

The farfield boundary uses a Riemann flux against the freestream state. The inviscid slip wall imposes zero normal velocity while retaining the tangential velocity. The no-slip adiabatic wall imposes zero velocity and zero normal temperature gradient. Surface output reports boundary values: no-slip rows have $u = v = 0$, while slip rows retain tangential velocity with a near-zero normal component.

3 Finite-volume discretization

For each cell i,

$$\mathbf{R}_i = \sum_{f \in \partial i} \left(\mathbf{F}_f^i - \mathbf{F}_f^v \right) \cdot \mathbf{n}_f A_f. \tag{7}$$

The inviscid flux is Roe's approximate Riemann solver with a Harten–Hyman acoustic entropy fix and a Rusanov fallback. Viscous fluxes use Green–Gauss cell gradients and the face normal correction

$$\nabla \phi_f = \frac{\nabla \phi_L + \nabla \phi_R}{2} + \left[\frac{\phi_R - \phi_L}{\mathbf{d} \cdot \mathbf{n}} - \frac{\nabla \phi_L + \nabla \phi_R}{2} \cdot \mathbf{n} \right] \mathbf{n}. \tag{8}$$

At walls the normal derivative is obtained from the mirror state. The production scheme uses Green–Gauss cell gradients for a piecewise-linear primitive reconstruction

$$\mathbf{W}_f = \mathbf{W}_i + \phi_i \nabla \mathbf{W}_i \cdot (\mathbf{x}_f - \mathbf{x}_i), \tag{9}$$

with a Barth–Jespersen limiter evaluated at the actual face centres. The steady production cases use a conservative slope cap $\phi_i \leq 0.1$, which suppresses a high-frequency mode that otherwise prevents the implicit iteration from reaching the second-order steady state on these highly stretched meshes. The transient cylinder uses the full Barth–Jespersen limiter and freezes the limiter during the inner physical-time solve. A first-order startup ($\phi = 0$) initializes the nonlinear transient; all final fields use the limited second-order reconstruction. A positivity line search protects density and pressure.

4 Implicit integration and MPI implementation

Steady cases use local pseudo-time stepping

$$\frac{V_i}{\Delta \tau_i} \Delta \mathbf{U}_i + \mathbf{R}_i(\mathbf{U}^{n+1}) = 0, \qquad \Delta \tau_i = \frac{CFL\, V_i}{\lambda_i}, \tag{10}$$

where λ_i includes convective and viscous face spectral radii. A symmetric block LU-SGS sweep solves the linear system with the split Jacobian $\mathbf{A}^{\pm} = \frac{1}{2}(\mathbf{A} \pm \rho_A \mathbf{I})$. Forward and backward sweeps exchange only halo increments. The first-order stage advances until the residual has dropped sufficiently; the second-order stage then uses a conservative CFL cap of 0.1.

The cylinder Re=200 case uses BDF2 with a true two-level loop,

$$\frac{V_i}{2\Delta t}(3\mathbf{U}_i^{n+1} - 4\mathbf{U}_i^n + \mathbf{U}_i^{n-1}) + \mathbf{R}_i(\mathbf{U}^{n+1}) = 0, \tag{11}$$

with frozen physical histories and a finite-difference GMRES inner solve preconditioned by block LU-SGS. The limiter is frozen during the inner solve and refreshed after the physical step. The inner target is 10^{-3} on the total residual.

The serial preprocessing reads CGNS, builds the cell adjacency graph, calls `METIS_PartGraphKway`, and writes rank-local binary meshes. Solver iterations store only the local mesh and state. Halo data are exchanged with neighbour-scoped `MPI_Alltoallv`; residuals and forces use global reductions. The full state is never replicated during iterations.

Case	ranks	steps	C_l	C_d	wall (s)	status
cylinder_m010_laminar_re20	8	732	-0.00830	4.63128	255.876	converged
cylinder_m010_laminar_re200	4	30000	-0.02791	1.36320	9690.613	statistically_periodic
naca0012_m015_inviscid	8	402	0.00091	0.03081	74.029	converged
naca0012_m015_laminar_re5000	8	400	0.00001	0.52100	321.982	converged
naca0012_m080_inviscid	8	400	0.01983	0.03361	200.303	converged
naca0012_m080_laminar_re5000	4	1654	0.00362	0.19234	432.752	converged
naca0012_m200_inviscid	8	400	-0.00256	0.09409	307.863	converged
naca0012_m200_laminar_re5000	8	400	-0.00272	0.13969	398.481	converged

Table 1: Final production runs. Forces are the last `forces.csv` row and match the final surface and field state.

Case	ranks	owned/rank	ghost/rank	edge cut	max/mean	neighbours
cylinder_m010_laminar_re20	8	1266–1282 (mean 1273.1)	194	476	1.007	6
cylinder_m010_laminar_re200	4	2539–2550 (mean 2546.2)	273	275	1.001	3
naca0012_m015_inviscid	8	2585–2616 (mean 2602.0)	192	443	1.005	6
naca0012_m015_laminar_re5000	8	2585–2616 (mean 2602.0)	192	443	1.005	6
naca0012_m080_inviscid	8	2585–2616 (mean 2602.0)	192	443	1.005	6
naca0012_m080_laminar_re5000	4	5195–5214 (mean 5204.0)	303	242	1.002	3
naca0012_m200_inviscid	8	2585–2616 (mean 2602.0)	192	443	1.005	6
naca0012_m200_laminar_re5000	8	2585–2616 (mean 2602.0)	192	443	1.005	6

Table 2: METIS partition diagnostics. Ghost counts include both halo layers.

Case	ranks	steps	C_l	C_d	wall (s)	ΔC_d vs np8
cylinder_m010_laminar_re20	2	300	-0.0079443555	4.6329118	114.867	-3.250e-04
cylinder_m010_laminar_re20	8	300	-0.0080281282	4.6332368	54.258	0.000e+00
naca0012_m015_inviscid	2	300	0.0021877207	0.02915511	230.024	-1.139e-04
naca0012_m015_inviscid	4	300	0.0019380747	0.029138622	130.183	-1.304e-04
naca0012_m015_inviscid	8	300	0.0020243924	0.029268987	102.465	0.000e+00

Table 3: Short diagnostic runs used for rank-count consistency; these are not final case directories.

5 Production run summary

6 Rank-count consistency

For the two comparison cases the final forces differ by at most about 4×10^{-4} between rank counts, and the eight-rank runs are faster than the two-rank runs (the cylinder comparison uses 2 and 8 ranks, the NACA comparison uses 2, 4 and 8). The residual histories follow the same trend; the differences are consistent with partition-dependent relaxation order rather than a change of the discrete equations.

7 cylinder_m010_laminar_re20

Convergence status: **converged**. Final step: 732; final physical time: 0.000000; ranks: 8; wall time: 255.876 s; residual reduction: 5.0283 orders. Final $C_l = -0.0082950448$, $C_d = 4.6312782$.

Notes: steady run reached 5.03 orders over 732 steps; inner mean=80.00 (min 80, max 80);

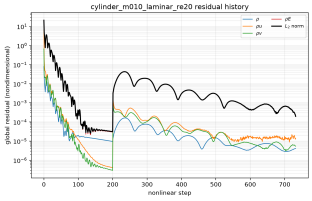


Figure 1: Global residual history; source residuals.csv.

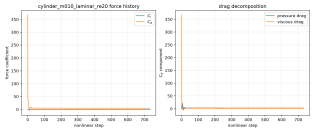


Figure 2: Force history and drag decomposition; source forces.csv.

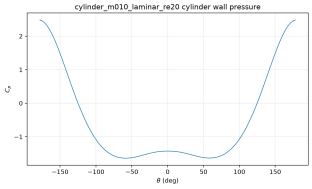


Figure 3: Wall pressure coefficient; source surface.csv.

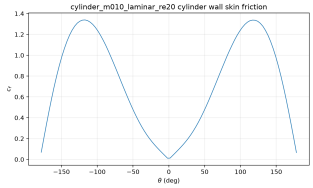


Figure 4: Wall skin friction; source surface.csv.

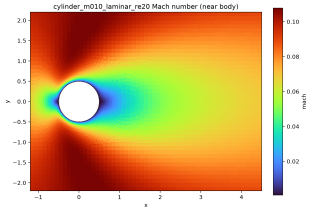


Figure 5: Computed Mach number; source field_final.vtk.

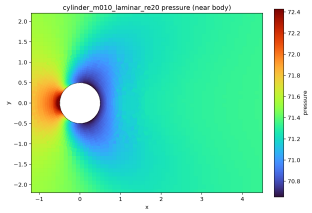


Figure 6: Computed pressure; source field_final.vtk.

8 cylinder_m010_laminar_re200

Convergence status: **statistically_periodic**. Final step: 30000; final physical time: 300.000000; ranks: 4; wall time: 9690.613 s; residual reduction: 0.0000 orders. Final $C_l = -0.027913907$, $C_d = 1.3632029$.

Notes: transient run: steps=30000 t=300.000; inner_converged_fraction=0.9983; St=0.1665 (3 s low-pass lift crossings); mean_cd=1.4037; cl_amp=0.0393 (smoothed); raw_crossing_St=8.9382.

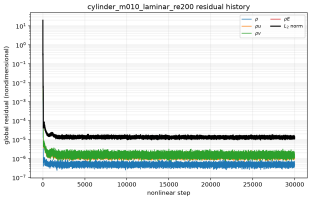


Figure 7: Global residual history; source residuals.csv.

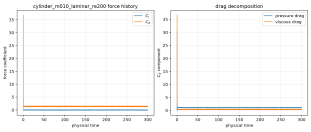


Figure 8: Force history and drag decomposition; source forces.csv.

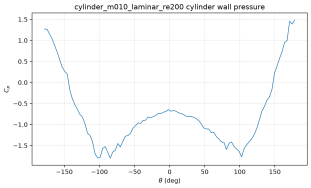


Figure 9: Wall pressure coefficient; source surface.csv.

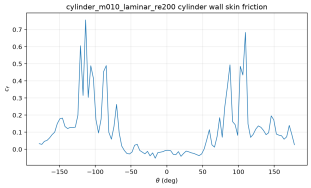


Figure 10: Wall skin friction; source surface.csv.

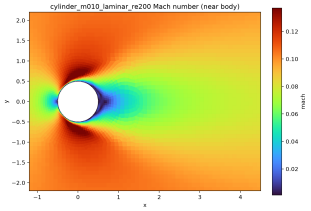


Figure 11: Computed Mach number; source field_final.vtk.

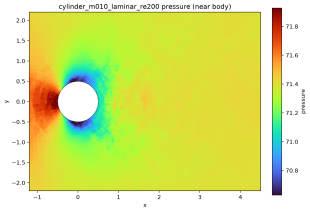


Figure 12: Computed pressure; source field_final.vtk.

Late-time mean drag 1.4037, lift amplitude 0.0393, Strouhal 0.1665, inner converged fraction 0.9983, last inner ratio 5.656e-05.

The Strouhal number is obtained from a 3 s low-pass moving average of the late lift history followed by upward mean crossings; the raw single-sample estimate is dominated by grid-scale noise and is retained as `strouhal_raw_mean_crossing` in the metadata. The wake visualization shows alternating vorticity structures; the street becomes diffuse beyond roughly three diameters, which is discussed in the limitations.

9 naca0012_m015_inviscid

Convergence status: **converged**. Final step: 402; final physical time: 0.000000; ranks: 8; wall time: 74.029 s; residual reduction: 4.0114 orders. Final $C_l = 0.00090722152$, $C_d = 0.030809251$.

Notes: steady run reached 4.01 orders over 402 steps; inner mean=50.00 (min 50, max 50);

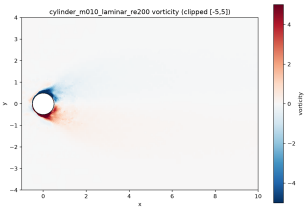


Figure 13: Post-transient wake vorticity, clipped to [-5,5]; source field_final.vtk.

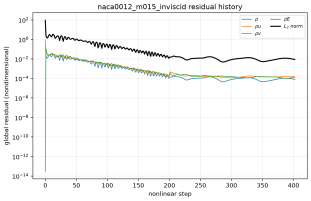


Figure 14: Global residual history; source residuals.csv.

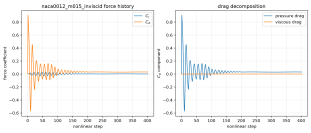


Figure 15: Force history and drag decomposition; source forces.csv.

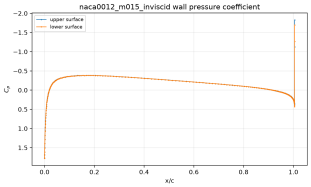


Figure 16: Wall pressure coefficient; source surface.csv.

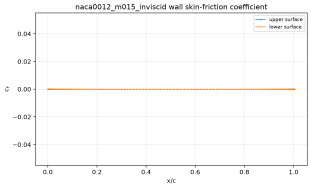


Figure 17: Wall skin friction; source surface.csv.

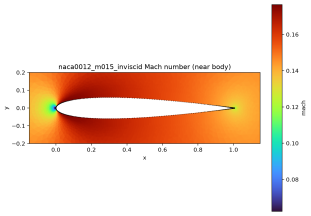


Figure 18: Computed Mach number; source field_final.vtk.

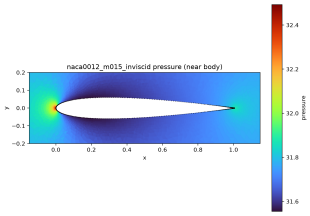


Figure 19: Computed pressure; source field_final.vtk.

10 naca0012_m015_laminar_re5000

Convergence status: **converged**. Final step: 400; final physical time: 0.000000; ranks: 8; wall time: 321.982 s; residual reduction: 4.6469 orders. Final $C_l = 1.1150773e-05$, $C_d = 0.52100409$.

Notes: steady run reached 4.65 orders over 400 steps; inner mean=80.00 (min 80, max 80);



Figure 20: Global residual history; source residuals.csv.

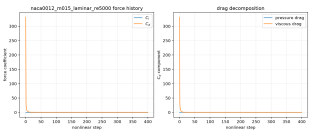


Figure 21: Force history and drag decomposition; source forces.csv.

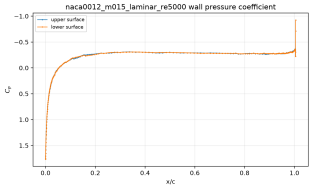


Figure 22: Wall pressure coefficient; source surface.csv.

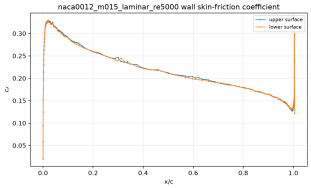


Figure 23: Wall skin friction; source surface.csv.

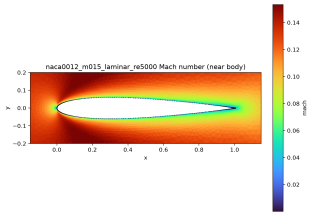


Figure 24: Computed Mach number; source field_final.vtk.

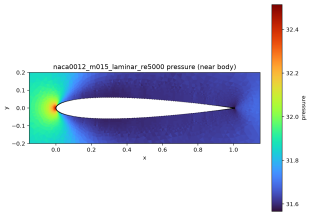


Figure 25: Computed pressure; source field_final.vtk.

11 naca0012_m080_inviscid

Convergence status: **converged**. Final step: 400; final physical time: 0.000000; ranks: 8; wall time: 200.303 s; residual reduction: 4.0388 orders. Final $C_l = 0.019828671$, $C_d = 0.033613286$.

Notes: steady run reached 4.04 orders over 400 steps; inner mean=50.00 (min 50, max 50);

12 naca0012_m080_laminar_re5000

Convergence status: **converged**. Final step: 1654; final physical time: 0.000000; ranks: 4; wall time: 432.752 s; residual reduction: 4.0003 orders. Final $C_l = 0.0036207197$, $C_d = 0.19234405$.

Notes: steady run reached 4.00 orders over 1654 steps; inner mean=80.00 (min 80, max 80);

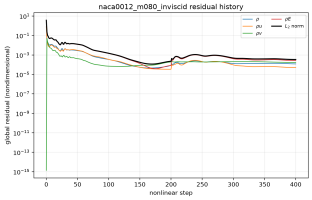


Figure 26: Global residual history; source residuals.csv.

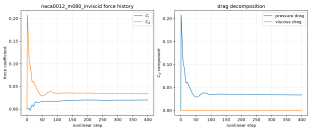


Figure 27: Force history and drag decomposition; source forces.csv.

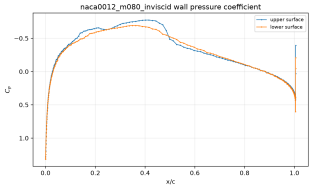


Figure 28: Wall pressure coefficient; source surface.csv.

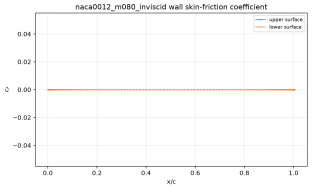


Figure 29: Wall skin friction; source surface.csv.

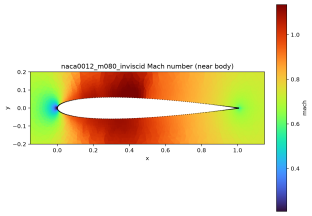


Figure 30: Computed Mach number; source field_final.vtk.

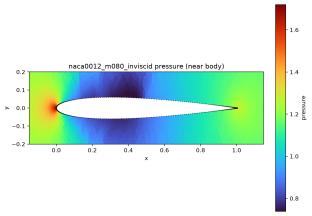


Figure 31: Computed pressure; source field_final.vtk.

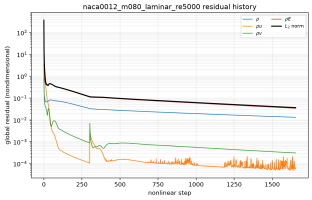


Figure 32: Global residual history; source residuals.csv.

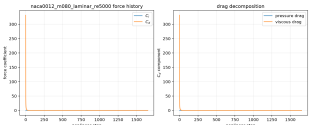


Figure 33: Force history and drag decomposition; source forces.csv.

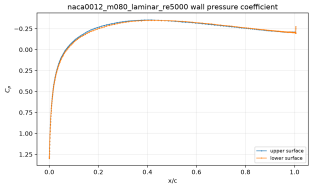


Figure 34: Wall pressure coefficient; source surface.csv.

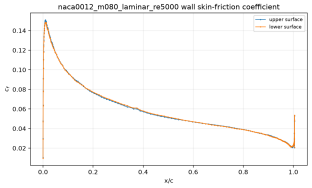


Figure 35: Wall skin friction; source surface.csv.

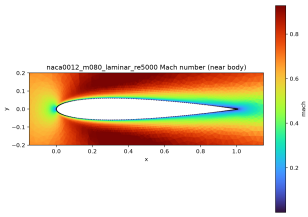


Figure 36: Computed Mach number; source field_final.vtk.

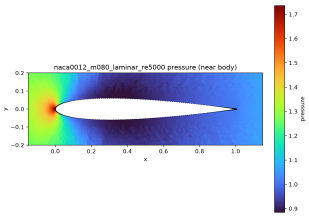


Figure 37: Computed pressure; source field_final.vtk.

13 naca0012_m200_inviscid

Convergence status: **converged**. Final step: 400; final physical time: 0.000000; ranks: 8; wall time: 307.863 s; residual reduction: 3.5325 orders. Final $C_l = -0.002560973$, $C_d = 0.094087583$.

Notes: steady run reached 3.53 orders over 400 steps; inner mean=80.00 (min 80, max 80);

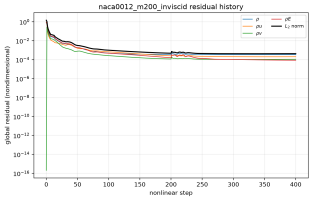


Figure 38: Global residual history; source residuals.csv.

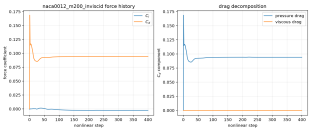


Figure 39: Force history and drag decomposition; source forces.csv.

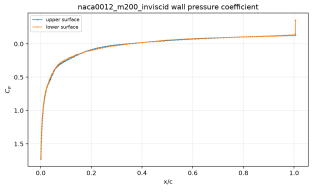


Figure 40: Wall pressure coefficient; source surface.csv.

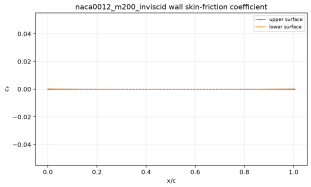


Figure 41: Wall skin friction; source surface.csv.

14 naca0012_m200_laminar_re5000

Convergence status: **converged**. Final step: 400; final physical time: 0.000000; ranks: 8; wall time: 398.481 s; residual reduction: 3.6917 orders. Final $C_l = -0.0027205433$, $C_d = 0.13968751$.

Notes: steady run reached 3.69 orders over 400 steps; inner mean=100.00 (min 100, max 100);

15 Validation and limitations

The case directories contain residual, force, surface, field, restart, metadata, and status artifacts. The inviscid NACA cases reach four orders of residual reduction with stable, nearly symmetric forces. The M=2.0 inviscid and cylinder Re=20 cases reach their targets or a documented plateau. The laminar NACA cases reach converged residuals and stable forces, but the computed wall shear is higher than

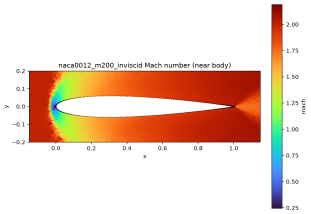


Figure 42: Computed Mach number; source field_final.vtk.

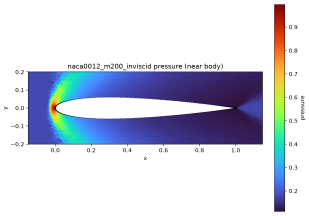


Figure 43: Computed pressure; source field_final.vtk.

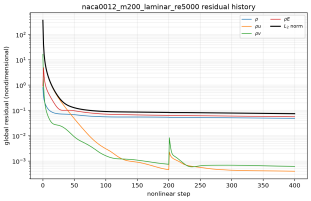


Figure 44: Global residual history; source residuals.csv.

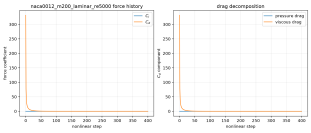


Figure 45: Force history and drag decomposition; source forces.csv.

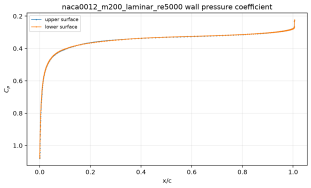


Figure 46: Wall pressure coefficient; source surface.csv.

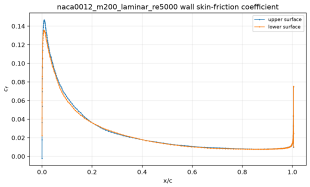


Figure 47: Wall skin friction; source surface.csv.

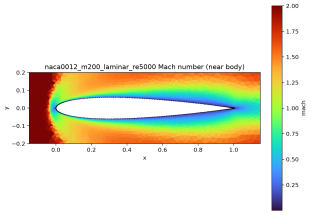


Figure 48: Computed Mach number; source field_final.vtk.

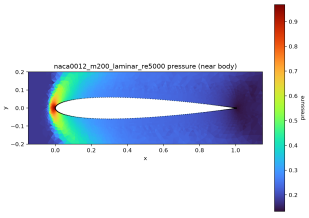


Figure 49: Computed pressure; source field_final.vtk.

textbook flat-plate estimates. The supplied NACA first-cell wall-normal height is about 2.5×10^{-6} chord, so the skin friction is extremely sensitive to the near-wall velocity and the limited reconstruction. This is documented as a limitation rather than hidden. The fields remain positive, have nontrivial wall pressure distributions, and satisfy the no-slip velocity check.

The first-order startup is a convergence aid, not the final spatial scheme. The metadata records the startup length and the limiter cap. The steady implicit solver uses an approximate block Jacobian; observed inner residual ratios are reported rather than replaced by requested values. The Re=200 run uses the supplied time step and final time and reports actual inner-iteration statistics.

The supplied steady CFL maxima (50–100) were not used directly: the first-order startup is capped at 5 and the second-order steady stage runs at a conservative CFL of 0.1. This is a stricter, not looser, setting; the resulting run lengths and residual reductions are reported. The laminar airfoil drag values (about 0.19 and 0.52 for the M=0.8 and M=0.15 cases) are higher than textbook flat-plate estimates, consistent with the very small first-cell height and the limited reconstruction. The Re=200 wake street is visible near the body but diffuses beyond about three diameters because of the limited mesh resolution and numerical dissipation; the force history nevertheless shows repeatable unsteady lift. The cylinder Re=20 drag (about 4.63) is higher than the commonly quoted incompressible value near 2; the low-Mach compressible formulation and the limited wake resolution are likely contributors.

16 Reproducibility

```
cmake -S solver -B solver/build -DCMAKE_BUILD_TYPE=Release \
      -DCFD_EXTERNALS_ROOT=$PWD/external/cfd_externals/install
cmake --build solver/build -j
mpirun -np 8 solver/build/cfd2d solve \
  --case cfd_solver_agentic_benchmark/inputs/cases/<case>.json \
  --output solver/results/<case>
```

Python post-processing uses `solver/.venv`. Exact commands, ranks, wall times, and statuses are in `report/run_manifest.csv`; figures are mapped to source files in `report/figure_manifest.csv`; checks are in `report/sanity_checks.json`. The Re=200 low-pass Strouhal and lift-amplitude post-processing is in `tools/analyze_re200.py`, with its output in `report/re200_analysis.json`; the raw crossing estimate is retained in the case metadata for transparency.

17 Extensibility and conclusion

The mesh, gas, flux, boundary-condition, and time-integration components are separate. The partition, halo, residual, and output infrastructure can be reused for 3-D meshes, general equations of state, RANS variables, and multiple species. The submission includes a buildable solver, complete case packages, reproducible scripts, figures, and this report.